

3.

| f  | AB | 00 | 01 | 11 | 10 |
|----|----|----|----|----|----|
| CD | 00 | 0  | 1  | 1  | 1  |
| 01 | 0  | 0  | 1  | 1  | 1  |
| 11 | 1  | 0  | 1  | 1  | 1  |
| 10 | 1  | 0  | 1  | 1  | 1  |

SOP: 找"1"

$$f(A,B,C,D)$$

$$= B\bar{C} + A + BC$$

POS: 找"0"

$$f(A,B,C,D) = (A+B+C)(A+\bar{B}+\bar{C})$$

| f  | AB | 00 | 01 | 11 | 10 |
|----|----|----|----|----|----|
| CD | 00 | 1  | 1  | 0  | 0  |
| 01 | 0  | 0  | 0  | 0  | 0  |
| 11 | 0  | 0  | 0  | 0  | 0  |
| 10 | 1  | 1  | 0  | 0  | 0  |

SOP

$$f(A,B,C,D)$$

$$= \bar{A}\bar{D}$$

POS

$$f(A,B,C,D)$$

$$= (\bar{A})(\bar{D}) = \bar{A}\bar{D}$$

| f | AB | 00 | 01 | 11 | 10 |
|---|----|----|----|----|----|
| C | 0  | 0  | 1  | 0  | 1  |
| 1 | 1  | 0  | 1  | 0  | 0  |

SOP

$$f(A,B,C,D)$$

$$= \bar{A}B\bar{C} + A\bar{B}\bar{C}$$

$$+ \bar{A}\bar{B}C + ABC$$

POS

$$f(A,B,C,D) = (A+B+C)(\bar{A}+\bar{B}+C)(A+\bar{B}+\bar{C})(\bar{A}+B+\bar{C})$$

4.

(a)

| f  | WX | 00 | 01 | 11 | 10 |
|----|----|----|----|----|----|
| YZ | 00 | 1  | 0  | 1  | 0  |
| 01 | X  | 1  | X  | 1  | X  |
| 11 | 1  | 0  | 1  | 0  | 0  |
| 10 | X  | 1  | X  | 1  | 1  |

显然, 把X看成1更容易化简

$$f(W,X,Y,Z) = \bar{W}\bar{X} + WX + \bar{Y}Z + Y\bar{Z}$$

(b)

| f  | AB | 00 | 01 | 11 | 10 |
|----|----|----|----|----|----|
| CD | 00 | 1  | 1  | 0  | 0  |
| 01 | 1  | X  | 0  | 0  | 0  |
| 11 | 0  | 0  | 0  | X  | 1  |
| 10 | 0  | 0  | 1  | 1  | 1  |

$$X = 1$$

$$f(A,B,C,D) = \bar{A}\bar{C} + AC$$

5. 无冒险. 指卡诺图中不存在相切的圈.

b)

| F  | AB | 00 | 01 | 11 | 10 |
|----|----|----|----|----|----|
| CD | 00 | 1  | 1  | 0  | 0  |
| 01 | 0  | 1  | 1  | 1  | 1  |
| 11 | 0  | 1  | 0  | 0  | 1  |
| 10 | 0  | 1  | 1  | 1  | 0  |

这里存在相切的情况, 如

一个冗余项().

相切, 加冗余项().

相切加冗余项

可以用SOP, 也可以用POS均可.

$$F(A,B,C,D) = \bar{A}\bar{C}\bar{D} + \bar{A}B + B\bar{C}D$$

$$+ A\bar{C}D + ABD + B\bar{C}\bar{D}$$

$$F(A,B,C,D) = (A+B+\bar{D})(A+B+\bar{C})$$

$$(B+\bar{C}+D)(\bar{A}+B+D)$$

$$(\bar{A}+C+D)(\bar{A}+\bar{B}+\bar{C}+\bar{D})$$

改造成或非

$$\overline{F(A,B,C,D)} = \overline{\bar{A}\bar{C}\bar{D} + \bar{A}B + B\bar{C}D + A\bar{C}D + ABD + B\bar{C}\bar{D}}$$

$$= \overline{\bar{A}\bar{C}\bar{D}} \cdot \overline{\bar{A}B} \cdot \overline{B\bar{C}D} \cdot \overline{A\bar{C}D} \cdot \overline{ABD} \cdot \overline{B\bar{C}\bar{D}}$$

$$= (A+C+D) \cdot (A+\bar{B}) \cdot (\bar{B}+C+\bar{D}) \cdot (\bar{A}+C+\bar{D})$$

$$(\bar{A}+B+\bar{D})(\bar{B}+\bar{C}+D)$$

$$= \overline{A+C+D} + \overline{A+\bar{B}} + \overline{\bar{B}+C+\bar{D}} + \overline{\bar{A}+C+\bar{D}}$$

$$+ \overline{\bar{A}+B+\bar{D}} + \overline{\bar{B}+\bar{C}+D}, \text{ 每一个小项可以用或非实现.}$$

各个小项作为最后大或非门的输入, 最后取反

即可. ( )

POS类似, 更简单一点, 直接展开即可.

(c)

| F | AB | 00 | 01 | 11 | 10 |
|---|----|----|----|----|----|
| C | 0  | 0  | 0  | 0  | 1  |
| 1 | 0  | 1  | 1  | 1  | 1  |

$$F(A,B,C) = \bar{A}\bar{B} + AC + BC$$

$$F(A,B,C) = (A+B)(A+C)(\bar{B}+C)$$